Report Narrative

What the Customer Behaviour dashboard says, page by page

Dataset	E-commerce customer behaviour: orders, product catalogue and customer engagement (3 linked tables)
Period / size	1 Jan 2023 – 26 Mar 2024 | 5,000 orders | 5,000 unique customers | 8 product categories | 10 cities
Report	Mini_project.pbix, 4 pages: Executive Overview, Customer Insights, Product, Profit & Delivery, Marketing & Engagement
Currency	Shown as ₺ (Turkish Lira), assumed from the Turkish cities in the data. Please confirm against the original source.

1. Purpose

This report turns raw order, product and engagement data into a story a business owner can act on. It answers four questions in order: **how is the business performing, who is buying, what is making money, and how do customers find and stay with us.** Each page of the Power BI dashboard answers one of these questions, and the narrative below follows the same order.

2. The story in brief

> The business generated **₺4.92M revenue from 5,000 orders** (average order value ₺983) and an estimated **₺1.36M gross profit** (27.7%). Growth is real but narrow: it comes from **bigger baskets, not more orders**, and it leans on **one category (Electronics, 47% of revenue)** and **one city (Istanbul, 27%)**. Customers are broadly satisfied (average rating 3.90 out of 5) but **every category sits below its target rating**. Marketing and loyalty tools exist but are not yet lifting order value: the loyalty programme and discounts show no basket-size benefit, and the highest-volume channel (Social Media) brings the lowest-value orders.

3. Page-by-page narrative

Page 1 – Executive Overview: how is the business doing?

The page opens with six headline cards: total revenue ₺4.92M, 5,000 orders, ₺983 average order value, ₺1.36M estimated gross profit, a 3.90 average rating and 60% returning customers. Below them, the category bar chart shows how concentrated sales are. **Electronics alone brings ₺2.33M (47%)**, followed by Home & Garden (₺0.91M) and Sports (₺0.75M); the remaining five categories together contribute under 20%. Interestingly, each category receives a similar number of orders (about 610–670), so the gap is explained by price level, not popularity: an Electronics order averages ₺3,732 against ₺118 for Books.

The city chart repeats the pattern geographically. **Istanbul contributes 27% of revenue and the top three cities (Istanbul, Ankara, Izmir) contribute 52%.** Smaller cities such as Kayseri (AOV ₺1,215) and Adana (₺1,130) spend more per order than Istanbul (₺1,039), which points to under-tapped potential. The monthly trend moves between ₺257K and ₺397K with peaks in August 2023, December 2023 and January 2024. March 2024 looks like a drop but is only a partial month because data stops on 26 March; the dashboard should be read with that caveat. On a like-for-like basis (first quarter 2023 versus the first 86 days of 2024), revenue grew about 7% (about 12% per day), with average order value up 10% while order count slipped 3%. Credit and debit cards account for 66% of orders, digital wallets 19%, bank transfer 10% and cash on delivery

just 5%.

Page 2 – Customer Insights: who is buying?

Revenue is concentrated in the **25–44 age bands, which generate 59% of sales**; the 18–24 group adds 20%, 45–54 adds 15% and the 55+ group only 5%. The smallest group is, however, the most valuable per order (₺1,028). Gender is almost evenly split (female 50%, male 48%) while the small "Other" group (1.5% of orders) has the lowest average order value (₺829). **Mobile dominates: 56% of orders and 57% of revenue**, with the highest AOV (₺1,006); tablet is the weakest (10% of orders, ₺921 AOV).

The heatmap of age group by category confirms that Electronics is the leading category in every age band (45–49% of each band's revenue), so age does not change what people buy, only how many of them buy. Returning customers make up 60% of orders but spend and rate the same as new customers (AOV ₺986 versus ₺979), meaning loyalty is not yet translating into higher value.

Page 3 – Product, Profit & Delivery: what is making money, and is it good enough?

Revenue and profit tell different stories. Electronics earns 47% of revenue but only **38% of gross profit** because its margin (22%) is among the lowest. Higher-margin categories (Beauty 42%, Fashion 38%, Toys 36%, Books 35%) are small. The category table also reveals that **all eight categories fall short of their target rating**. Food has the widest gap (3.83 against a 4.5 target, −0.67), followed by Beauty (−0.49) and Sports (−0.40); Fashion is closest to target (−0.12). The price-versus-rating scatter shows that price level has almost no effect on satisfaction.

Two operational charts add nuance. **Discounts** are applied to 30% of orders and total ₺124K (2.5% of gross sales), yet discounted and non-discounted orders have the same average value (₺981 versus ₺984). **Delivery** averages 6.5 days; only 19% of orders arrive in 1–3 days and 32% take eight days or longer. Fast delivery earns a slightly higher rating (3.95 versus about 3.89), but the effect is small, so speed alone will not close the rating gap.

Page 4 – Marketing & Engagement: how do customers arrive and stay?

Search Engine is the strongest channel: 22% of revenue from 19% of orders, with the highest AOV (₺1,108). **Social Media brings the most orders (22%) but the lowest AOV (₺926) and the lowest satisfaction score (3.66)**. Referral orders are large (₺1,026) but carry the lowest rating (3.83). Cart abandonment averages 1.39 per customer (6,940 in total) and is highest for Paid Ads (1.49) and Referral (1.48).

On engagement, 57% of customers open marketing email and those who do spend about 4% more per order. Email is also the most preferred contact method (35% of orders, AOV ₺1,038), while **20% of customers have no contact preference recorded**. The loyalty programme enrols 37% of customers, yet **660 members (35% of members) hold zero points**, and members spend 12% less per order than non-members (₺908 versus ₺1,027). Overall customer satisfaction is 3.72 out of 5; 16% of customers score 1 or 2.

4. How the four pages connect

Question	Page	Finding	Implication
How are we doing?	Executive Overview	Growth comes from basket size; sales depend on Electronics and Istanbul	Diversify categories and cities
Who buys?	Customer Insights	25–44 and mobile users drive sales; returning = new in value	Protect the core, add value for returning customers
What pays?	Product, Profit & Delivery	Profit mix differs from revenue mix; all categories miss target rating	Shift mix to margin, fix quality
How do they stay?	Marketing & Engagement	Channels differ in order value; loyalty and discounts do not lift baskets	Re-balance channels, redesign loyalty

5. Data notes and limitations

- **Partial month:** the data ends on 26 March 2024, so March 2024 and any full-year 2024 comparison are incomplete. Year-over-year views should use matched periods.
- **One order per customer:** each of the 5,000 customers appears once, so repeat-purchase frequency and customer lifetime value cannot be measured. “Returning” is a flag in the data, not an observed repeat order.
- **Estimated gross profit:** calculated as order amount × the category’s estimated margin from the product catalogue. It is not an accounting profit figure (no cost of goods, shipping or returns data).
- **Currency:** assumed to be Turkish Lira (₺) from the city names. Verify against the original dataset source.
- **Gauge targets:** the targets shown on the Marketing & Engagement gauges are benchmark values set by the analyst, not company targets.
- **Weak relationships:** ratings, satisfaction, delivery time, cart abandonment and order value show almost no correlation with each other in this dataset (all below ±0.02). Findings should therefore be read as descriptive patterns, not proof of cause and effect.
- **Missing values:** “Preferred contact method” is blank for 1,002 customers (20%), shown as “Not Specified” in analysis.